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Application of the FFT in classifying building energy change point models

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Contents

1 Abstract **2**

2 Background **2**
2.1 Introduction . . . 2
2.2 Problem . . . 3
2.3 Solution . . . 3

3 Technical Overview **4**
3.1 The Fast Fourier Transform . . . 4
3.1.1 Direct Derivation . . . 4
3.1.2 The Goertzel FFT Algorithm . . . 5
3.2 XGBoost . . . 6
3.2.1 Decision Tree Ensembles . . . 6

4 Data and Analysis **6**
4.1 Direct FFT . . . 6
4.2 Goertzel FFT . . . 8

5 Conclusions **9**

6 References **10**

List of Figures

1 Map of U.S. climate zones obtained from Pacific Northwest National Laboratory. . . . 2
2 Sample 3PC, 4P, and 5P time series (left) and corresponding usage plotted against OAT (right). . . . 3
3 Comparison of real and simulated utility data models: 4P (left) and 5P (right). . . . 4
4 Comparison of frequency spectra with $f_s = 24$ samples/period: $N = 24$ (left) and $N = 32$ (right). . . . 7
5 Accuracy of classifier trained on time series and frequency spectra for 4P vs. 5P. . . . 7

6 Accuracy of classifier trained on time series and frequency spectra for 4P vs. 3PC. 8
7 Frequency spectrum obtained from Goertzel FFT algorithm. 9
8 Accuracy of classifier trained on time series and Goertzel FFT spectra. 9

1 Abstract

The northeastern United States has one of the most variable climates in the world, with wide temperature swings over the course the of the year. This variation largely influences how energy is consumed in urban systems [2,5]. Building energy consumption is often analyzed with respect to monthly average outside air temperature (OAT). Using monthly utility data and recorded OAT, piecewise linear regression models can be generated to determine particular performance metrics of buildings. These regression models, though, are statistically bottlenecked in their ability to reliably classify building performance, because they depend on the location of a computed change point [2]. The ensuing report details a continued effort to combat these statistical ambiguities by applying frequency transformations to 24-month utility time series data and leveraging periodicity in annual energy consumption to effectively classify change point models. The direct and Goertzel fast Fourier transform (FFT) algorithms were applied to the discrete-time electricity consumption sequences, and machine learning (ML) models based on gradient-boosted decision tree ensemble techniques were trained on: a) The 24-month time series data for each class of change point model; b) 24-point direct FFT data describing the time series waveforms in each class; c) 32-point direct FFT data describing the time series waveforms zero-padded to a sequence of 32; and d) 32-point Goertzel FFT outputs. Performance was compared between the ML systems trained on utility data and the models produced using FFT methods. By developing an efficient classification system centered around digital frequencies in the time series samples, the need for iterative regression fitting and statistical decision making is eliminated. In other words, new 24-point sequences can be classified accordingly, before running a single regression model for further analysis.

2 Background

2.1 Introduction

Buildings are responsible for a large portion of nationwide energy consumption. Estimates for residential and commercial building energy use are between 30-40% of all energy consumed across the nation; and upwards of 70% of all electricity [1,3]. These records make building energy use a target area for developing and optimizing energy efficient strategies and technologies. Regions of high climate variability are subject to influence by wide temperature swings and humidity. Pacific Northwest National Library (PNNL) constructed a climate zone map highlighting areas of similar climate around the continental United States (Fig. 1)

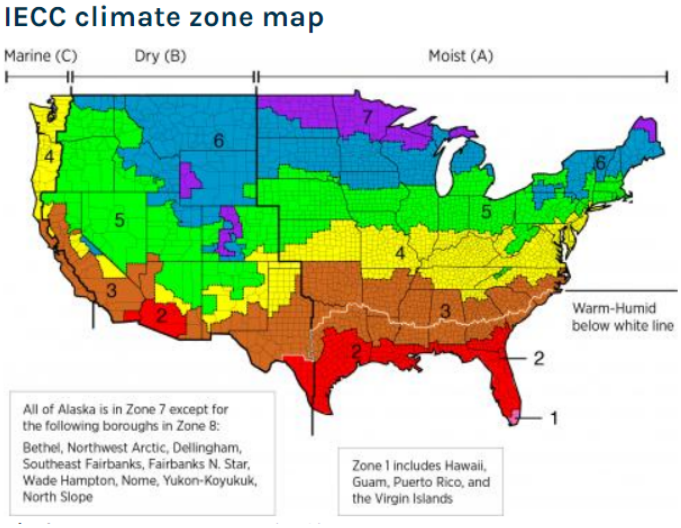


Figure 1: Map of U.S. climate zones obtained from Pacific Northwest National Laboratory.

The climate in the northeast United States undergoes a high degree of fluctuation. The air around the Mid-Atlantic states is humid and temperatures in a given year can range from above 100°F to below 0°F. Generally electricity use spikes in the summer months, when the outside air temperature is high and systems need to be cooled; whereas, fuel use rises in the winter months to keep indoor temperatures comfortably warm. As a means of optimizing building operations, energy data is plotted against average monthly OAT to generate an associated change point model. Figure 2 illustrates sample 3PC, 4P, and 5P time series of electricity consumption (left) and the corresponding plots of usage against OAT (right).

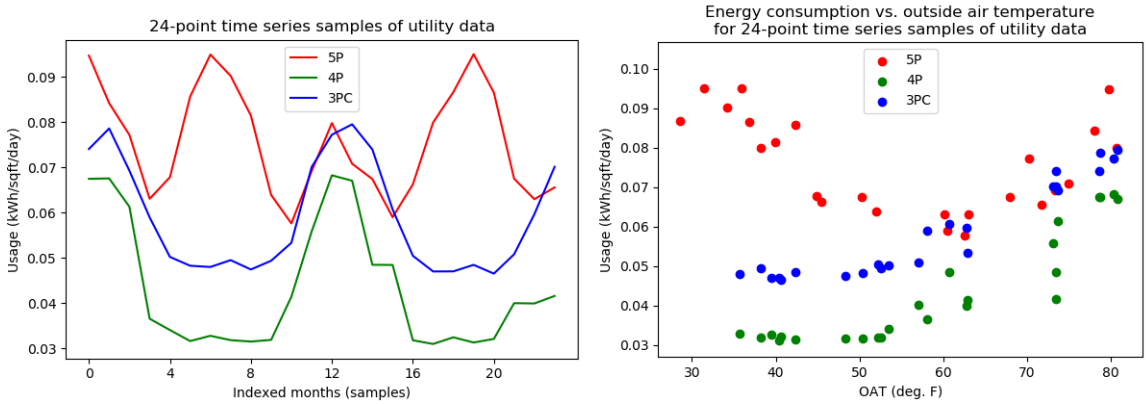


Figure 2: Sample 3PC, 4P, and 5P time series (left) and corresponding usage plotted against OAT (right).

2.2 Problem

New York City is located in Climate Zone 4A, which is host to several types of change point models, including the examples provided in Fig. 2. To optimize building operations, a facility is designated a best-fit change point model based on its energy consumption data with respect to OAT. In real statistical analysis, though, this task becomes difficult, because a certain level of tolerance is required to account for embedded noise. That is, randomness within the data presents some variance that may skew the best-fit change point model in favor of one class against another, resulting in a model misclassification [4]. Such clearly observable similarities are highlighted in Fig. 2, where the 3PC and 4PE models appear to be close translations of one another.

To determine a model that accurately fits a particular building, an algorithm would typically run piecewise linear regressions on the 24-point utility data for 5 change point model classes (3PC/H, 4PE/H, 5P). Once regression models of a sequence are obtained, decisions regarding the "best fit" model must be made on the basis of statistical benchmarks, like R^2 and CVRMSE. This decision making process leads to the following question: If two samples are regressed and yield tightly fit models with comparable statistical characteristics, which model *truly* represents the building dynamics more accurately?

2.3 Solution

While change point models provide some means of further understanding building system performance, they come at the cost of bias and loss of information. That is, certain parameters of these piecewise linear regressions depend on the computed values of other regression parameters. Specifically, aspects of heating and cooling sensitivity (slopes of the regression model) rely on the location of a calculated change point, which in effect is influenced by the data variance. Rather than giving attention to declaring what the best model *is* for a given system, it may prove useful to first focus on determining what is *not* the best model.

This project entails the development of a preliminary classification system to performing piecewise linear regressions on building utility time series. The classification system leverages periodicity in the data associated with seasonal temperature swings, to ultimately distinguish change point models by the frequency components present in the discrete-time waveform. After a model is classified by its frequency components, regression and analysis can ensue accordingly; rather than running 5 piecewise regressions and statistically splitting hairs on the best fit before gaining any real insight. Two FFT systems were employed, where the results of direct FFT computations and the Goertzel FFT algorithm were applied

to gradient boosted decision trees to perform classification on hundreds of 3PC, 4PE, and 5P change point models.

3 Technical Overview

From observation of the time series in Fig. 2, a broad model of the building dynamics can be derived, where the signal is decomposed into DC, AC, and noise components; in the case of a building, these features would more accurately represent baseload, seasonal oscillations in energy consumption, and random noise, respectively. Mathematically, this model can be represented by

$$E(t) = A_0 + Bg(t) + n(t), \tag{1}$$

where A_0 indicates the DC amplitude; B is a constant; $g(t)$ is a periodic function (i.e. sinusoidal of some frequency); and $n(t)$ is random noise. Figure 3 highlights simulated discrete-time utility data generated using the model described in eqn.(1).

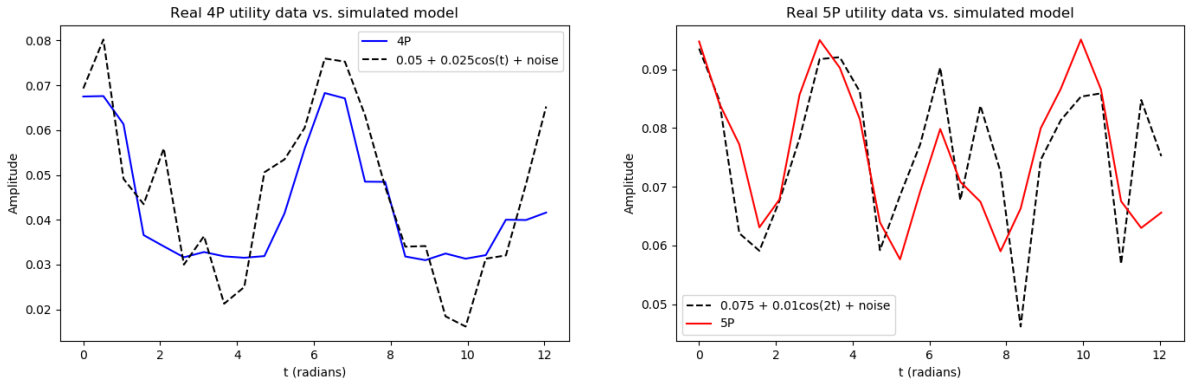


Figure 3: Comparison of real and simulated utility data models: 4P (left) and 5P (right).

Periodicity for a discrete-time signal is dependent on the number of points in the signal. Specifically, the fundamental frequency of a discrete-time waveform can be determined by

$$\omega_o N = 2\pi k \tag{2}$$

where ω_o is the fundamental frequency, N is the number of points in the sequence, and $k = 1, 2, 3, \ldots$. By fitting the annual 12-month cycle to a period of 2π, the real utility data was mapped across 2π radians and overlaid with the simulated waveforms. The result is a strong match between simulated and real utility data. For the sake of analysis, the overall sampling period will be considered within the full 24-point sequence, that is, the sampling frequency $f_s = 24$ samples/period. Considering this defined sampling period and frequency, it is suggested from Fig. 3 that a 5P model ought to have a different harmonic composition than a 4P or 3PC model. The indication of different frequency components may be key in quickly determining whether or not a set of data should even be fit to a particular change point model.

3.1 The Fast Fourier Transform

The data displayed in Fig. 3 suggests that a frequency domain approach to analysis could prove useful in determining one change point model against another prior to performing any regressions. To access the harmonic composition of the data, FFT algorithms were implemented. The frequency data obtained using direct and Goertzel FFT methods were then split into training and testing data sets, and applied to a gradient-boosted decision tree model (XGBoost) for classification.

3.1.1 Direct Derivation

Much of the frequency domain analysis takes into account that the functions are complex. While the sequence may be of real data, the discrete Fourier Transform (DFT) leverages Euler's relation of complex

exponentials to define a windowing function

$$W_N^{kN} = e^{j\frac{2\pi}{N}Nk} = e^{j2\pi k} = cos(2\pi k) + jsin(2\pi k) = 1, k = 0, 1, 2, \ldots, N-1, \tag{3}$$

where N is the number samples, i.e. the window length. Using this windowing function, the discrete Fourier Transform (DFT) can be obtained by

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}, k = 0, 1, \ldots N-1 \tag{4}$$

Direct computation of the DFT requires the decomposition of $x(n)$ and W_N into real and imaginary parts before carrying out any multiplication. Expanding eqn.(4) to account for real and imaginary aspects yields the following:

$$X(k) = \sum_{n=0}^{N-1} \left[\left(Re\{x(n)\} Re\{W_N^{kn}\} - Im\{x(n)\} Im\{W_N^{kn}\} \right) + j \left(Re\{x(n)\} Im\{W_N^{kn}\} + Im\{x(n)\} Re\{W_N^{kn}\} \right) \right] \tag{5}$$

with $k = k = 0, 1, \ldots N-1$. From inspection of eqn.(5), each kth term in the sequence demands $4N$ real multiplications and $4N-2$ real additions. Thus, the total DFT sequence mandates $4N^2$ real multiplications and $N(4N-2)$ real additions when $k \to N-1$. However, the periodicity and symmetry of the windowing function in eqn.(3) can be exploited to reduce the computational expense.

For a (right-sided) sequence of length N to be periodic, it must repeat in value every N samples. Applying this principle to W_N^{kn}, we see that

$$W_N^{kn} = W_N^{k(n+N)} = W_N^{(k+N)n},$$

which holds true for integer multiples of N. Additionally, we note the function is conjugate symmetric in that,

$$W_N^{k(N-n)} = W_N^{kn*}$$

Combining the properties and making a substitution of variables, eqn.(5) can be rewritten as

$$X(k) = W_N^{-kN} \sum_{r=0}^{N-1} x(r) W_N^{kr} = \sum_{r=0}^{N-1} x(r) W_N^{-k(N-r)} = y_k(N), \tag{6}$$

where $y_k(n)$ is the result of input sequence $x(n)$ convolved with the windowed unit impulse response. This application of the Fourier Transform reduces the number of real and complex multiplications to about N^2, introducing the FFT.

3.1.2 The Goertzel FFT Algorithm

While the FFT greatly reduces the number of computations required to translate discrete-time signals to the frequency domain, the concept has been expanded upon to generate alternative algorithms. One such method is the Goertzel FFT. The Goertzel FFT utilizes the Z-transform (ZT) to map discrete-time data to digital frequency spectra.

What makes the frequency domain favorable is that, convolved time signals can be manipulated using simple multiplication, reducing the amount of time required for computation. Despite the similarities between the real utility data and the simulated models in Fig.3, it is unconfirmed that the waveform is indeed a periodic signal. Rather, what is known, is that the real utility data is represented by a real, right-sided sequence of shifted impulse samples. From this information, the ZT of $x(n)$ can be computed by

$$X(z) = Z\{x(n)\} = \sum_{k=0}^{N-1} x(k) z^{-k} = x(0)z^0 + x(1)z^{-1} + x(2)z^{-2} + \cdots + x(N-1)z^{-(N-1)} \tag{7}$$

where $z = e^{j\frac{2\pi k}{N}}$. Because the input sequence has been digitized, the transfer function of the unit impulse response must also be converted to Z domain. In doing so, the first-order transfer function of the unit impulse response is found to be

$$H(z) = \frac{1}{1 - W_N^{-k} z^{-1}}, \tag{8}$$

and this can be multiplied to the Z-domain input sequence $X(z)$ to yield the output sequence:

$$Y(z) = H(z)X(z) \tag{9}$$

3.2 XGBoost

As mentioned, the system developed in this project serves the purpose of classifying change point models based on the frequency composition of discrete-time signals. After computing frequency data using FFT algorithms, these data samples were fed to a supervised machine learning model that operates on the basis of decision tree ensembles. This supervised learning method uses sets of classification and regression trees to determine the probable class of a given input feature.

3.2.1 Decision Tree Ensembles

The classification and regression trees (CARTs) with decision tree ensembles are different from traditional decision tree implementations in that the leaves only provide decision values [8]. These regression values establish a foreground for model optimization. In a classification problem, the prediction is made from input data, which varies based on the task at hand. The model is shaped to find the best-fitting parameters for the training data and respective class labels using an objective function, which is a means of measuring and optimizing model performance.

In defining the objective function, we consider the loss encountered in the training process and a regularization term. This objective function can be generalized to

$$obj(\theta) = L(\theta) + \Omega(\theta) \tag{10}$$

where $L(\theta)$ and $\Omega(\theta)$ represent the loss and regularization functions, respectively. The loss function can take on a number of different forms. Depending on the objective task, the loss function can be computed using mean squared error, or binary or multinomial cross-entropy. In this case, the loss function takes the form of softmax regression, which computes cross-entropy. In the case of 2 classes, softmax regression calculates the binary cross-entropy:

$$J = \frac{-1}{M}\sum_{i=1}^{M}[y_i log(p(y_i)) + (1-y_i)log(1-p(y_i))]$$

where M is the total number of samples; y_i is the ith sample target value and $p(y_i)$ is the probability of ith sample assuming the target value. This loss function was decided on due to the nonlinear nature of building systems and its functionality in binary classification [7,8].

The regularization function determines the complexity of the tree. The tree $f(x)$, as written by the developers, is a function of the scores on leaves ω. Then the regularization term can be described by

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda\sum_{j=1}^{T}\omega_j^2 \tag{11}$$

where T is the number of leaves on the trees and γ is the learning coefficient. Then eqn.(10) becomes

$$obj(\theta) = \frac{-1}{M}\sum_{i=0}^{M}[y_i log(p(y_i)) + (1-y_i)log(1-p(y_i))] + \gamma T + \frac{1}{2}\lambda\sum_{j=1}^{T}\omega_j^2 \tag{12}$$

4 Data and Analysis

4.1 Direct FFT

First, the direct FFT was applied to the utility data. Initially, sampling rate was set to 24 samples/period with $N = 24$ samples, before padding the sequence to $N = 32$ samples for 5P, 4P, and 3PC change point models. Figure 4 illustrates the resulting frequency spectra of the time series represented in Fig. 3 using the original signal of $N = 24$ samples (left) and the padded signal of $N = 32$ samples (right).

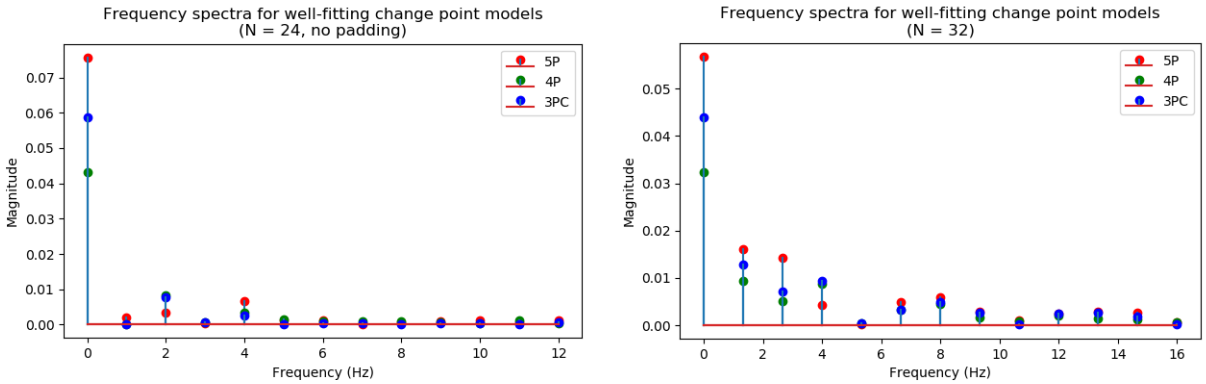


Figure 4: Comparison of frequency spectra with $f_s = 24$ samples/period: $N = 24$ (left) and $N = 32$ (right).

The unpadded system clearly indicates key frequencies at the fundamental 0 Hz, with additional standout components at 2 and 4 Hz. In this case, it can be seen that the 5P model retains a higher magnitude at 4 Hz, while the 3PC and 4P models show greater aspect at 2 Hz. It is important to note, however, that the difference in magnitude between these frequencies is on the order of 10^{-3} (i.e. the data closely emulate each other). When the sequence is zero-padded to a length of 32 samples, more frequencies show presence in the signal and the resolution changes. Rather than showing components at clean 1-Hz intervals, the spectrum has shifted to highlight frequencies in intervals of 1.33 Hz; the additional third comes from 8 zero-padded samples ($8/24 = 1/3$). Despite the adjusted frequency resolution, the spectra of each change point model is seen to be more distinct in geometry. The difference in the harmonic amplitudes is greater, and there are more frequencies of interest. The padded waveforms seem to take on a more identifiable shape in the frequency domain than the unpadded data.

Because there is such a notable difference between the padded and unpadded information, a comparison was made of the performance between the XGBoost model trained on the unpadded data, with XGBoost systems trained on the padded data. Additionally, these results were weighed against the data collected when training the system on time series samples. Figure 5 shows the classifier accuracy when trained on test sizes of 0.1, 0.2, 0.3, 0.4, and 0.5. In each split of training and testing data, the testing set was divided into 2 halves: The first half was kept for testing, while the second half established a validation set. The accuracy of predictions made on testing and validation sets were averaged to generate the accuracy scores observed in Fig. 5. Overall, the XGBoost sytem demonstrates significantly higher accuracy scores when trained on the frequency data, rather than the time series.

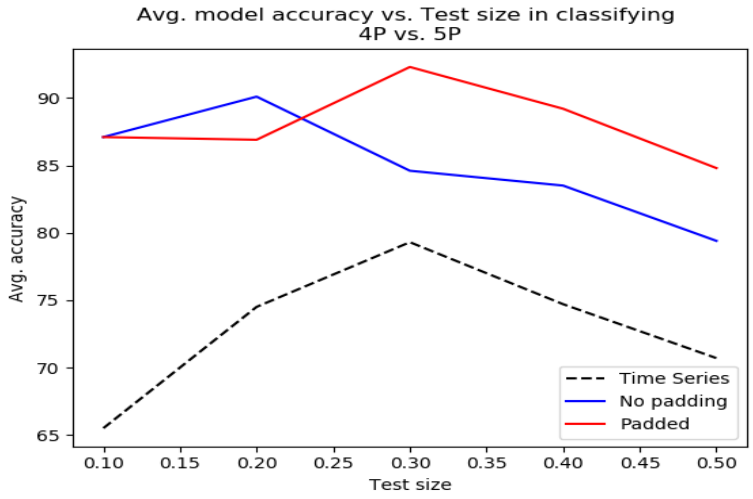


Figure 5: Accuracy of classifier trained on time series and frequency spectra for 4P vs. 5P.

When testing on 10% of the data set, the models trained on frequency data are more than 20% more accurate than the classifier trained on the time series. Generally, it appears that a testing size of 30% is optimal for this classification model. While the time series data tested an averaged best of 79.3%

accurate, the data padded to a length of 32 samples produced an averaged accuracy of 92.3%. Moreover, the padded sequences tested around 5-10% more accurate than the original 24-sample sequences. No changes were made to the classification model to generate higher accuracy scores.

For further analysis, 4P models were also tested against 3PC models. The accuracy scores generated for test sizes 0.1, 0.2, 0.3, 0.4, and 0.5 are shown in Fig. 6.

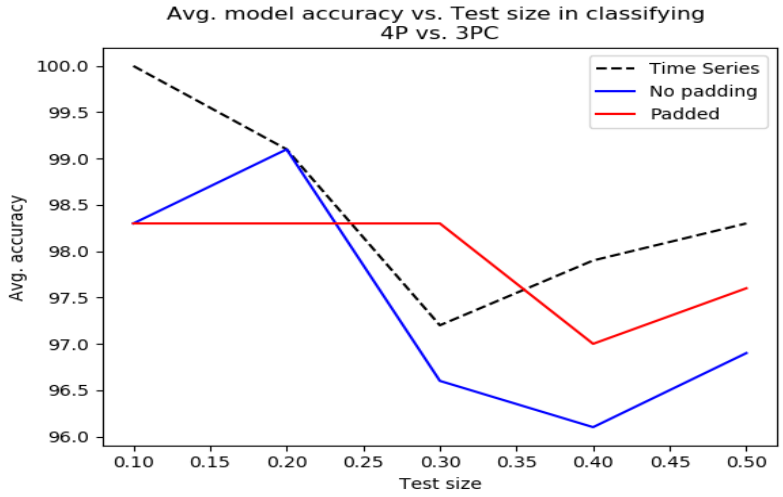


Figure 6: Accuracy of classifier trained on time series and frequency spectra for 4P vs. 3PC.

The accuracy data collected for the 4P/3PC classification is less conclusive in terms of general performance. While the accuracies for each of the time series and frequency data are all very high, and within 2% of each other for each test size, the most notable observation is the model sensitivity to size of the test data. That is, the accuracy of the time series and unpadded sequence models varied more with respect to testing ratio than that seen using padded sequences. Although the accuracy shows little difference between data formats, the classification results suggest a much stronger success rate when comparing 4P data with that of 3PC change point models. This may be due to the two-slope nature of the 4P model. Whereas 3-parameter cooling models hold a steady, flat line of energy consumption at low temperatures, 4-parameter models boast two distinct slopes, which may aid in the classification process. In other words, despite the observable geometry of the time series data, the features between 3PC and 4P models are more separable than those in comparing 4P and 5P change point models.

4.2 Goertzel FFT

Considering the substantially better classification performance put out by training and testing on frequency spectra, the Goertzel FFT was applied after padding the data to 32 samples and defining the new sampling frequency to be 32 samples/period. Resulting frequency spectra are seen in Fig. 7. The spectra collected using the Goertzel FFT algorithm are similar to those calculated using the direct method, but there are some notable differences. Using the Goertzel FFT, the spectra begin with the fundamental at 1 Hz, rather than at 0. The frequency components are also found at distinct integer values, with some aliasing occurring between the harmonics. In comparison, the direct method exhibited no aliasing when padded to 32 points, showing distinguishable lobes despite the shift in the location of frequency aspects due to the added zero-padded data points.

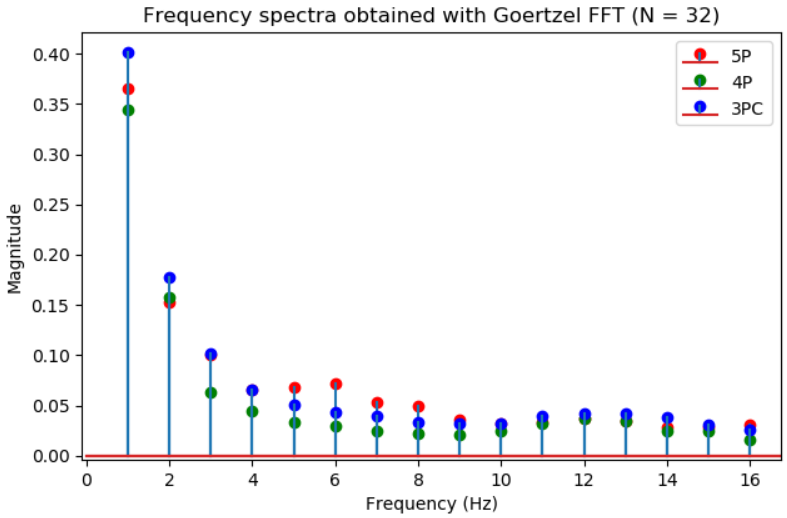


Figure 7: Frequency spectrum obtained from Goertzel FFT algorithm.

Figure 8 shows the model performance when using the Goertzel algorithm to obtain frequency spectra. Comparing this information to that in Fig. 5 and 6, the model accuracy when classifying 4P vs. 3PC with direct and Goertzel FFT methods are close together; there is extremely high accuracy, and minimal difference in accuracy when classifying with time series versus frequency data. However, there is substantial contrast in performance when classifying 4P against 5P data. When comparing 4P vs. 5P models, the direct FFT proves to classify the data roughly 20% more accurately than those trained on Goertzel spectra, regardless of the training-testing data ratio. This contrast in performance may be indicative of the effects of aliasing on identifying distinct signals.

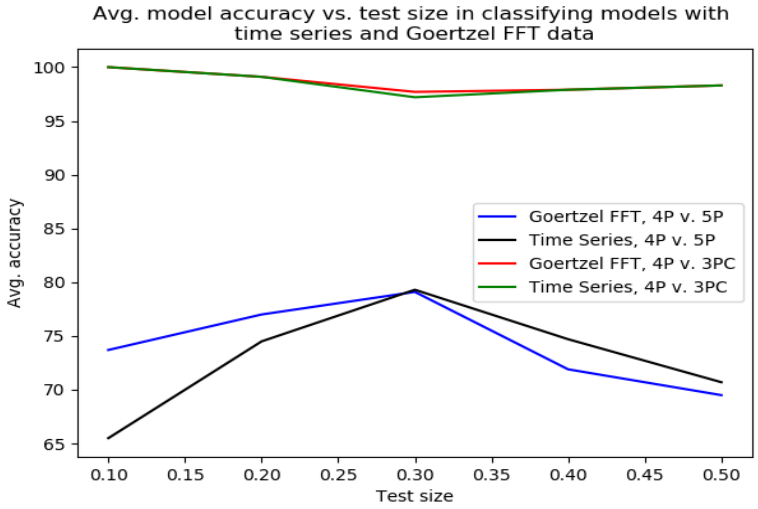


Figure 8: Accuracy of classifier trained on time series and Goertzel FFT spectra.

5 Conclusions

Promising solution The results collected suggest that the frequency domain proves useful in distinguishing 5P systems from other change point models. In other words, applying ML techniques for pre-regression classification would effectively reduce the amount of computing necessary to determine an appropriate change point model, rather than producing multiple ambiguous regressions and deciding which is best on a sole basis of statistical constraints. In addition to reducing the computational demand, the ML model can be expanded from binary to multi-class classification. In such a model, input sequences would be automatically classified as 3PH, 3PC, 4PE, 4PH, or 5P, before piecewise regression is performed

for analysis. This project highlights a proof-of-concept for a more robust system implementation.

Addressing performance issues While the classification system generally performed well, the results reflect a raw employment. No tuning or further optimization took place. Additionally, only the magnitudes of the frequency spectra were used in the classification process. In future cases, the data can be split into real and imaginary parts to more finely identify differences between various time series sets. A phase component can also be analyzed. Combining all of these aspects (real/imaginary parts, magnitude, and phase) can more closely dictate what makes each change point model distinct. Perhaps the use of such features can shed light on less obvious details of building dynamics. With building dynamics in mind, this system could potentially be expanded to sustain computation and classification on more granular interval data describing both the building as a whole, as well as individual subsystems within buildings [5].

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